

Introduction to Cloud Computing

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Agenda

- Migrating from On-Prem to Cloud
 - Cloud migration strategies
 - Cloud migration phases
- What is DevOps?
- DevOps Mindset & Principles
- The DevOps lifecycle
- Benefits of DevOps
- DevOps tools

Migrating from on-Prem to cloud



Cloud migration strategies

- **Rehost ("Lift and Shift")**

Applications, data, schema, and workloads are moved from the data center to an IaaS cloud deployment, without being changed in any way.

- **Refactor**

Redesign app to be cloud-native.

- **Replatform**

Minor cloud optimizations, e.g., switching database engine.

Cloud migration strategies cont'd

- **Replace(Repurchase)**

Move to SaaS

- **Retire**

Shut down unused resources.

- **Retain**

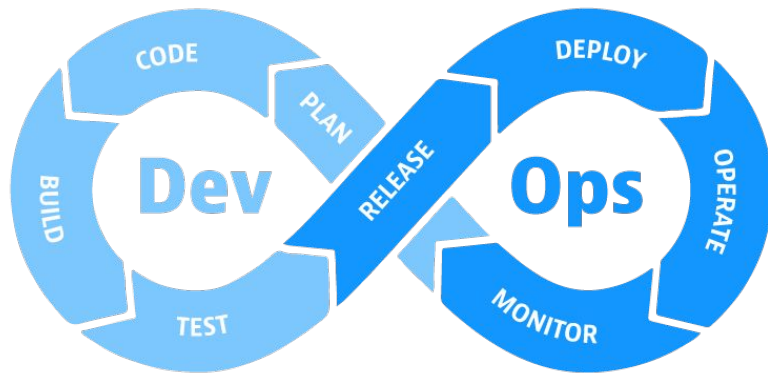
Keep on-prem temporarily or permanently for valid reasons.

Cloud migration phases

- Prepare
- Plan
- Migrate
- Operate
- Optimize

What is DevOps?

DevOps is the combination of cultural philosophies, practices, and tools that increases an organization's ability to deliver applications and services at high velocity.



DevOps Mindset & Principles

- **Collaboration:** Breaking down silos between Devs and Ops
- **Automation:** Doing less manual work
- **Continuous Feedback:** Learning fast, fixing faster
- **Small, frequent changes:** Safer deployments

Automation

Definition: Using tools to perform repetitive tasks automatically.

Examples in DevOps:

- Building code automatically (CI tools)
- Running automated tests
- Deploying to servers

Why it matters:

- Saves time
- Reduces human error
- Ensures consistent results

Monitoring

Definition: Continuously observing systems to detect issues before they become problems.

Examples:

- Application performance monitoring
- Server and network monitoring
- Logging and alerting

Why it matters:

- Catch errors early
- Maintain uptime & reliability
- Improve user experience

The DevOps lifecycle

- **Plan:** Requirements, user stories
- **Develop:** Writing code collaboratively
- **Build:** Integrating code into shared branches
- **Test:** Automated and manual testing for quality
- **Release:** Preparing for production
- **Deploy:** Getting code into production safely
- **Operate:** Keeping the system healthy
- **Monitor:** Tracking performance, errors, and user feedback

DevOps Vs. Traditional IT

	DevOps	Traditional IT
Collaboration	High: Cross-functional teams, shared ownership, real-time communication	Low: Siloed teams, limited communication, handoffs
Automation	High: CI/CD pipelines, Infrastructure as Code, automated testing	Low: Manual processes, manual deployments, manual configurations
Release Frequency	High: Frequent, small releases, continuous delivery	Low: Infrequent, large releases, batched deployments
Flexibility	High: Agile, adaptable to change, quick iterations	Low: Rigid processes, slow to adapt, limited flexibility

Benefits of DevOps

- **Speed**
- **Rapid Delivery**
- **Reliability**
- **Scale**
- **Improved Collaboration**
- **Security**

Core Services

Category	Tools
Version Control	Git, GitHub, Bitbucket
CI/CD	Jenkins, GitHub Actions
IaC	Terraform
Containerization	Docker, Kubernetes
Monitoring	Prometheus, Grafana

Thank You